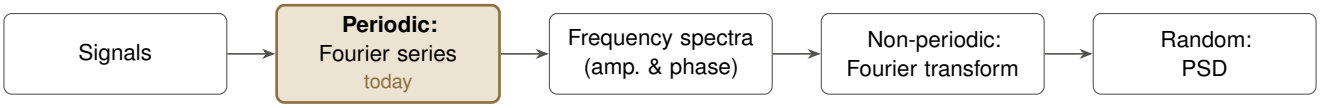
Spectral Analysis 1: Fourier Series



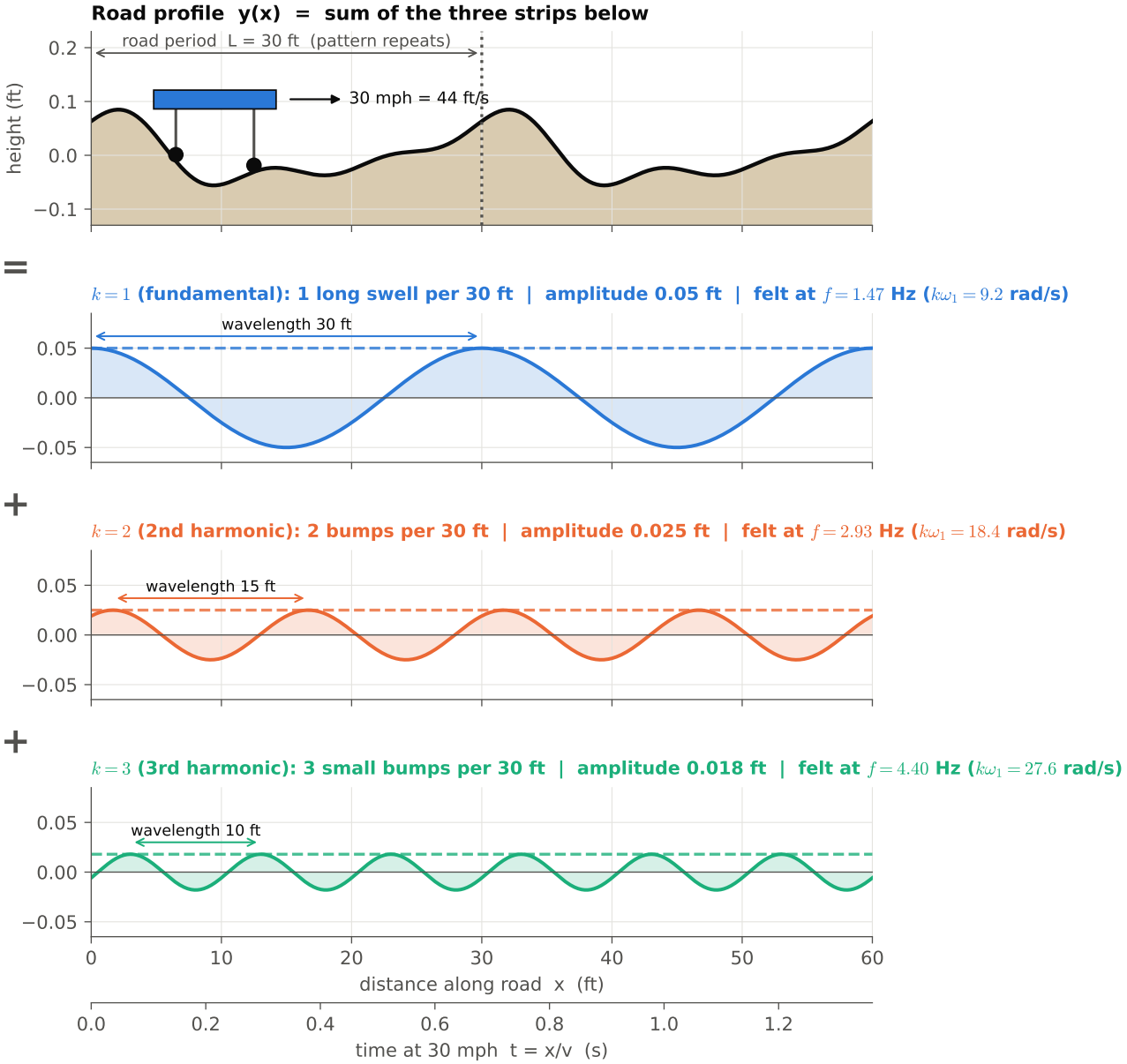
1 Why break signals into sinusoids?

We already know how an LTI system $G(s)$ responds to *one* sinusoid. A real input (such as a road profile) is not one sinusoid. Two facts let us handle it anyway:

- **Superposition:** $a\,u_1 + b\,u_2 \;\mapsto\; a\,y_1 + b\,y_2$.
- **Frequency response:** $M\cos(\omega t + \phi) \;\mapsto\; M\,|G(j\omega)|\cos\big(\omega t + \phi + \angle G(j\omega)\big)$ in steady state.

If $u(t) = \frac{A_0}{2} + \sum_k M_k \cos(k\omega_1 t - \theta_k)$, then the steady-state output is

$$y_{ss}(t) = \tfrac{A_0}{2}\,G(0) + \sum_{k=1}^{\infty} M_k\,|G(jk\omega_1)|\cos\big(k\omega_1 t - \theta_k + \angle G(jk\omega_1)\big).$$



The road (top) is exactly the sum of the three colored strips. Each strip is one *harmonic*: a perfectly sinusoidal road whose wavelength is L/k (30, 15, 10 ft). The car drives over each strip at 44 ft/s, so every strip shakes it at a single frequency $k\omega_1$, which is a sinusoidal input we already know how to handle with $G(j\omega)$. Vertical scale exaggerated.

How does a road wavelength turn into a frequency the car feels?

> The car crosses one wavelength L/k in $(L/k)/v$ seconds, so $f_k = k\,v/L$ and $\omega_k = k\omega_1$, with $\omega_1 = 2\pi v/L = 2\pi(44)/30 = 9.22$ rad/s. Faster driving or shorter bumps $\Rightarrow$ higher frequency.

Signal type	Tool	Spectrum
Periodic	Fourier **series**	lines at $k\omega_1$ only
Non-periodic (well-defined)	Fourier **transform**	continuous in ω
Random	Power spectral density	continuous, statistical

2 The Fourier series

For $x(t+T) = x(t)$, with fundamental frequency $\omega_1 = 2\pi/T$ rad/s:

$$x(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} \Big[A_k \cos(k\omega_1 t) + B_k \sin(k\omega_1 t) \Big]$$

- $k\omega_1$ is the kth harmonic.
- $A_0/2$ is the DC (average) value.

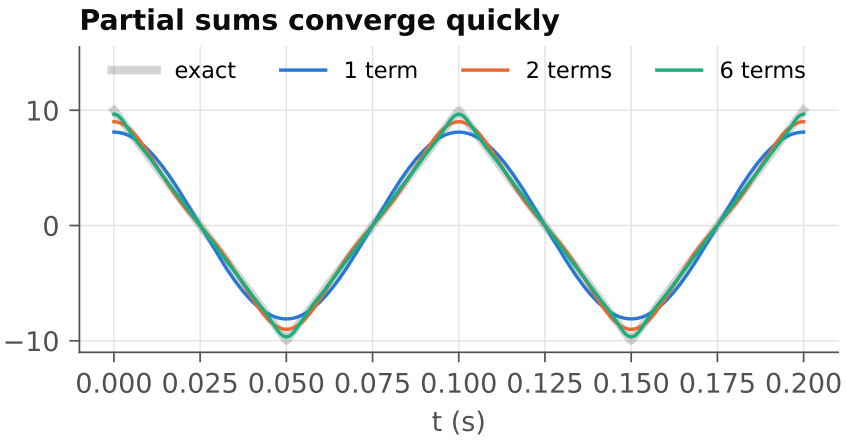
$$A_0 = \frac{2}{T}\int_0^T x(t)\,dt, \qquad A_k = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(k\omega_1 t)\,dt, \qquad B_k = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(k\omega_1 t)\,dt$$

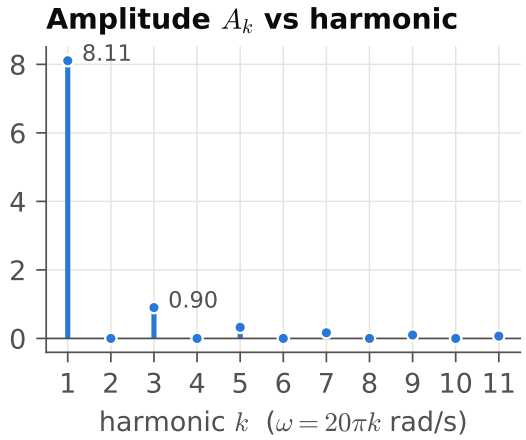
A_0 is just A_k at $k = 0$, which is why the DC term is written $A_0/2$.

2.1 Example: triangle wave (amplitude A, period T, even, zero mean)

$x(t) = \sum_{k=1}^{\infty} \frac{8A}{\pi^2(2k-1)^2}\cos\left((2k-1)\frac{2\pi}{T}t\right)$. With $A = 10$, $T = 0.1$ s:

$$x(t) \approx 8.105\cos(20\pi t) + 0.901\cos(60\pi t) + 0.324\cos(100\pi t) + 0.165\cos(140\pi t) + 0.100\cos(180\pi t) + 0.067\cos(220\pi t)$$

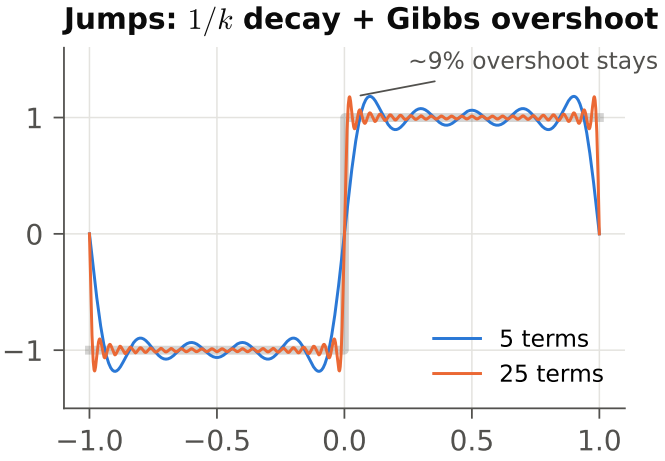




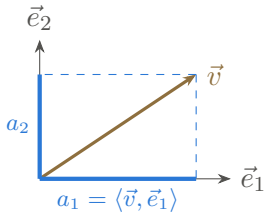
What do you notice about the harmonics?

Only **odd** harmonics appear (half-wave symmetry, $x(t + T/2) = -x(t)$). The amplitudes decay like $1/k^2$, so one term already gives 81% of the peak. Extra terms just sharpen the corners.

Smoothness sets the decay. A continuous signal with corners (triangle) has $|A_k| \sim 1/k^2$. A signal with jumps (square wave) has only $\sim 1/k$, so it needs many more terms. At a jump the series converges to the midpoint and keeps a ≈9% overshoot (Gibbs phenomenon).

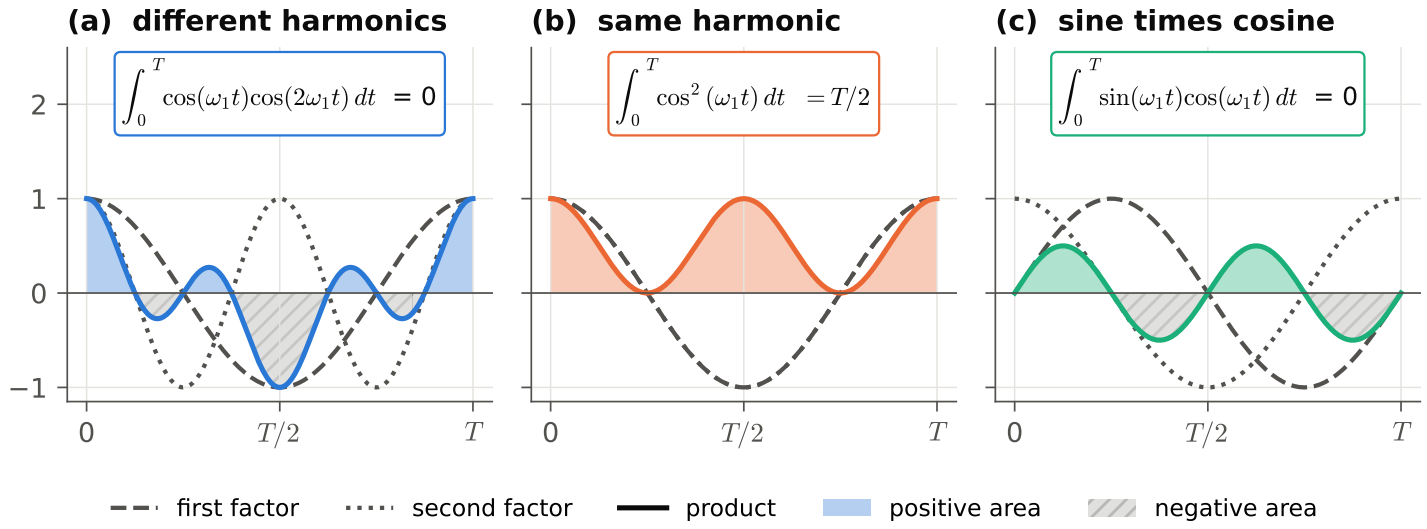


3 Geometric view: coefficients are projections



Vectors	Periodic functions
$\vec{v}$	$x(t)$
basis $\vec{e}_i$	$1,\ \cos k\omega_1 t,\ \sin k\omega_1 t$
$\langle \vec{v}, \vec{w} \rangle = \sum v_i w_i$	$\langle f, g \rangle = \int_0^T f\, g\, dt$
component a_i	coefficient A_k, B_k

To pull out one coefficient, say A_k, we multiply $x(t)$ by $\cos(k\omega_1 t)$ and integrate over one period. But $x(t)$ is itself a sum of harmonics (its Fourier series), so $x(t)\cos(k\omega_1 t)$ splits into one product per term of the series: $\cos(k\omega_1 t)$ times each harmonic in the series. Every one of those products is one of three kinds:

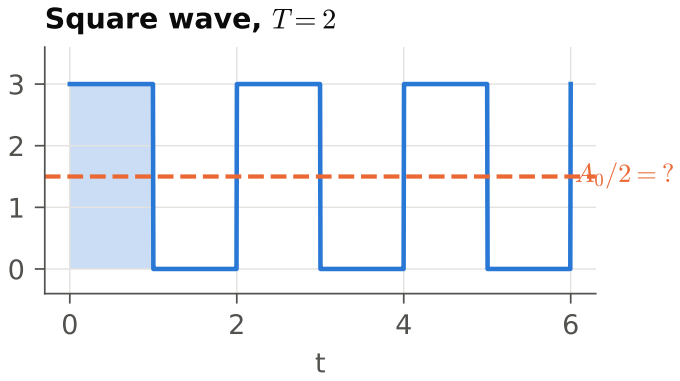


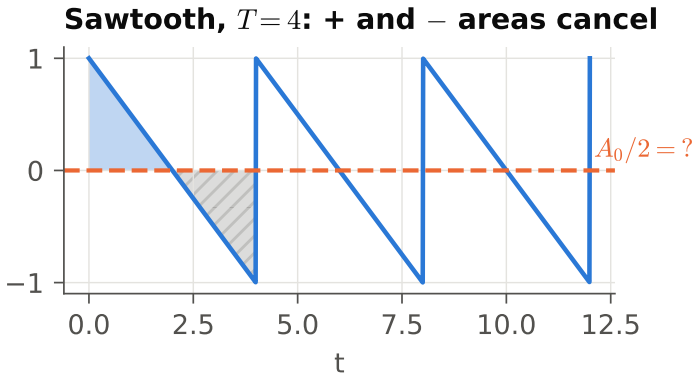
In (a) and (c) the positive and negative areas cancel, so the integral is zero: the two functions are *orthogonal*. In (b) the product $\cos^2$ is never negative, so an area of $T/2$ survives.

Takeaway. So $\int_0^T x(t)\cos(k\omega_1 t)\, dt$ keeps only the k-th harmonic: it equals $A_k \cdot T/2$, which gives $A_k = \dfrac{2}{T}\int_0^T x(t)\cos(k\omega_1 t)\, dt$.

4 Computing coefficients efficiently

4.1 DC term = average value



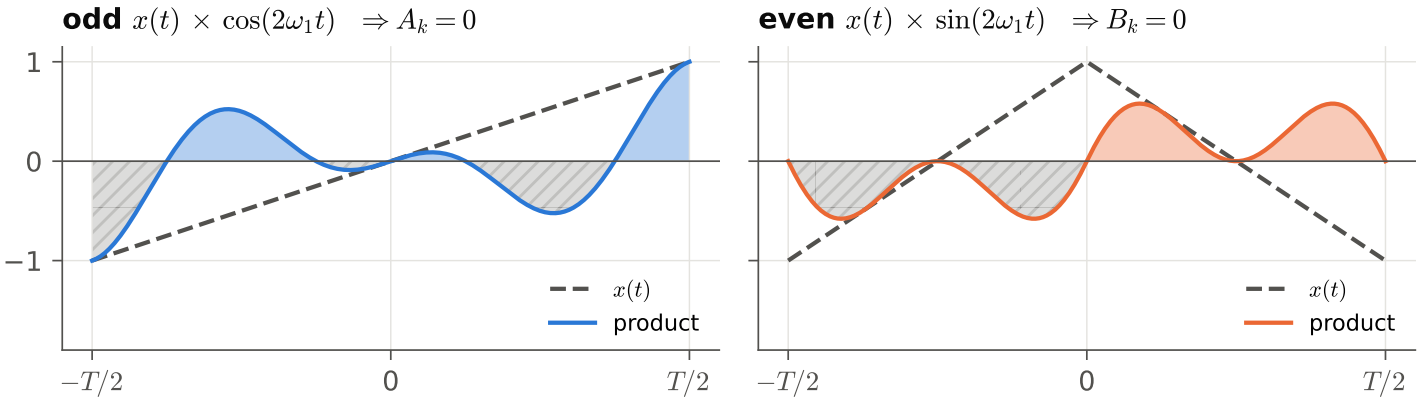


Square wave: $\dfrac{A_0}{2} = \dfrac{1}{T}\int_0^T x\, dt = \frac{1}{2}(3 \cdot 1) = \frac{3}{2}$

Sawtooth: $\dfrac{A_0}{2} = 0$ (areas cancel)

Any full period works: $\int_{t_0}^{t_0+T}$ gives the same result for every t_0, because the integrand is T-periodic. Pick the limits that make the algebra easiest, often $[-T/2,\ T/2]$.

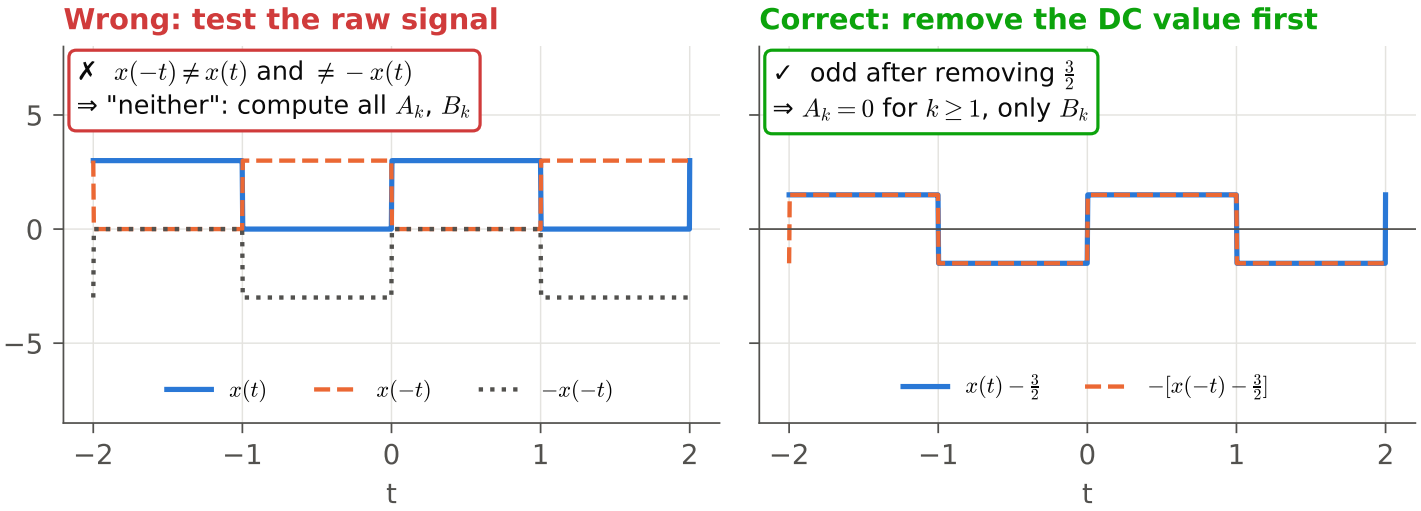
4.2 Symmetry shortcuts



odd × even = odd, and an odd function integrates to zero over $[-T/2, T/2]$.

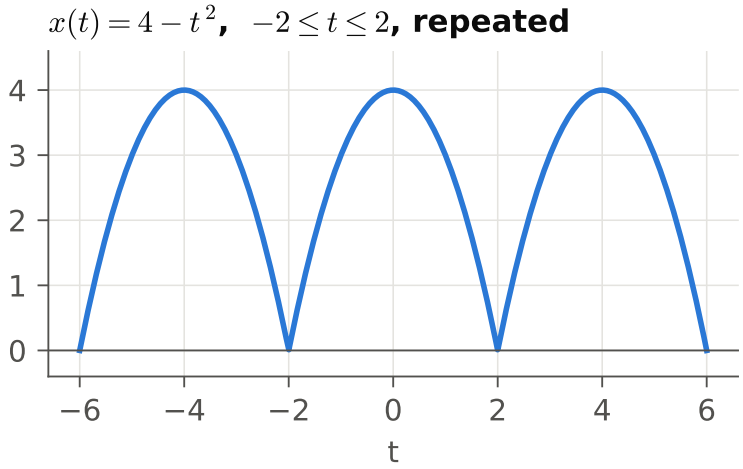
Symmetry	Test	Vanishing terms	Remaining integral
Odd	$x(-t) = -x(t)$	$A_k = 0$ (all k)	$B_k = \frac{4}{T}\int_0^{T/2} x \sin(k\omega_1 t)\, dt$
Even	$x(-t) = x(t)$	$B_k = 0$	$A_k = \frac{4}{T}\int_0^{T/2} x \cos(k\omega_1 t)\, dt$
Half-wave	$x(t + \frac{T}{2}) = -x(t)$	even harmonics	only odd k survive

Watch out. Symmetry depends on where you put $t = 0$, and can appear only after you subtract the DC value. An even signal may still have a DC term. Most signals are neither odd nor even, so expect both A_k and B_k.



The 0–3 square wave from Section 4.1. Tested as is, it looks like “neither” (left). Subtract its mean $3/2$ first and it is odd (right), so only B_k terms survive, plus the DC term $3/2$.

5 Worked example 1: periodic parabola



Period $T = 4$

$\omega_1 = 2\pi/T = \pi/2$ rad/s

Symmetry: even $\Rightarrow B_k = 0$

DC term

$$\frac{A_0}{2} = \frac{1}{4}\int_{-2}^{2}(4 - t^2)\,dt = \frac{1}{4}\left[4t - \tfrac{t^3}{3}\right]_{-2}^{2} = \frac{8}{3}$$

Cosine coefficients ($\omega_1 = \pi/2$, so $k\omega_1 t = k\pi t/2$)

Because $x(t)$ is even, $A_k = \dfrac{2}{4}\displaystyle\int_{-2}^{2}(4 - t^2)\cos\!\left(\tfrac{k\pi t}{2}\right)dt = \int_{0}^{2}(4 - t^2)\cos\!\left(\tfrac{k\pi t}{2}\right)dt.$

First part: $\displaystyle\int_0^2 4\cos\!\left(\tfrac{k\pi t}{2}\right)dt = \frac{8}{k\pi}\sin(k\pi) = 0.$

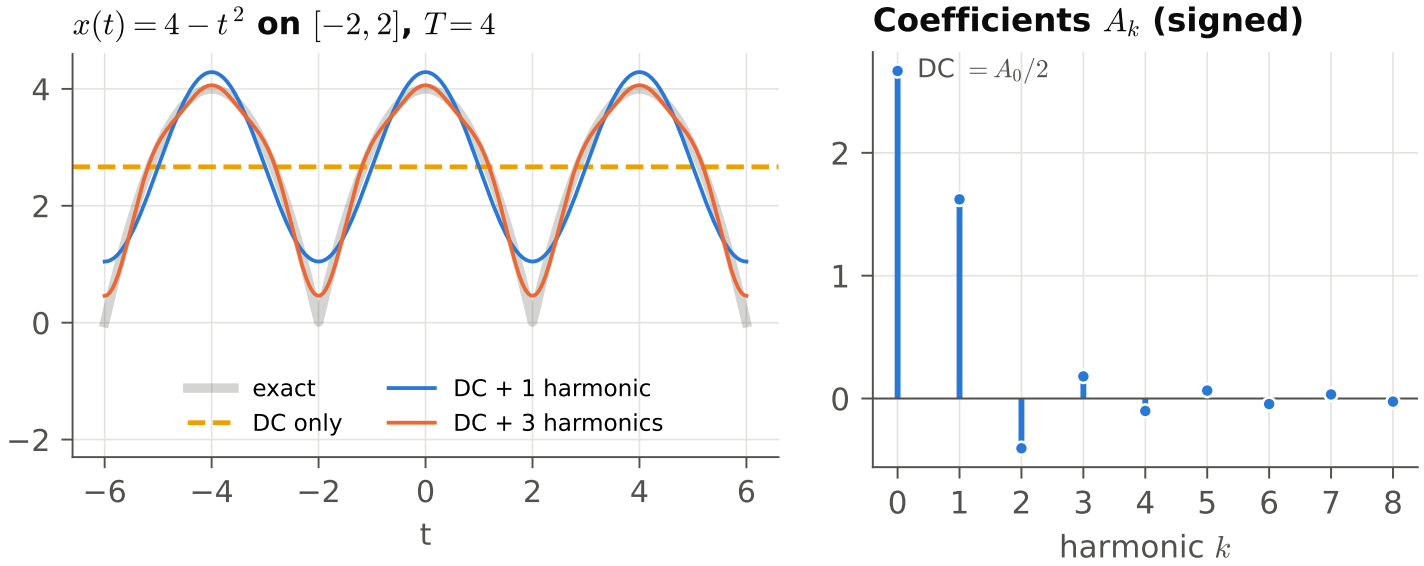
Second part (integrate by parts twice):

$$\int_0^2 t^2\cos\!\left(\tfrac{k\pi t}{2}\right)dt = \left[\tfrac{2t^2}{k\pi}\sin\tfrac{k\pi t}{2} + \tfrac{8t}{(k\pi)^2}\cos\tfrac{k\pi t}{2} - \tfrac{16}{(k\pi)^3}\sin\tfrac{k\pi t}{2}\right]_0^2 = \frac{16\cos(k\pi)}{(k\pi)^2} = \frac{16(-1)^k}{(k\pi)^2}$$

$$\Rightarrow \quad A_k = 0 - \frac{16(-1)^k}{(k\pi)^2} = \frac{16}{(k\pi)^2}(-1)^{k+1}$$

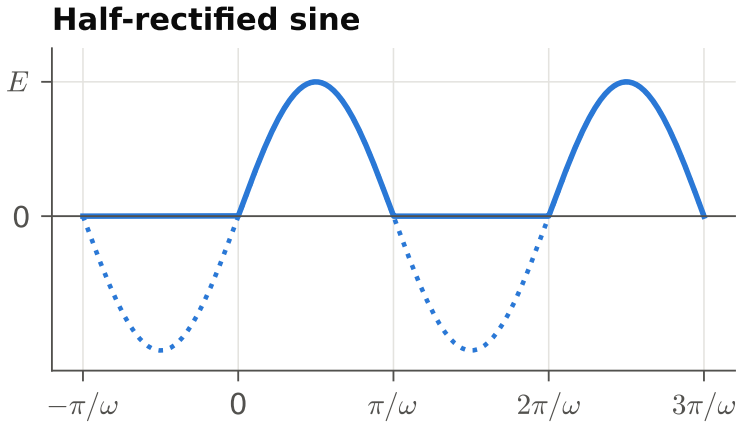
Series

$x(t) = \dfrac{8}{3} + \dfrac{16}{\pi^2}\left(\cos\tfrac{\pi t}{2} - \tfrac{1}{2^2}\cos\pi t + \tfrac{1}{3^2}\cos\tfrac{3\pi t}{2} - \tfrac{1}{4^2}\cos 2\pi t + \cdots\right)$. Check at $t = 0$: $\tfrac{8}{3} + \tfrac{16}{\pi^2}\cdot\tfrac{\pi^2}{12} = 4$ ✓



Left: partial sums. Right: signed coefficients. All harmonics appear, with alternating sign, decaying like $1/k^2$.

6 Worked example 2: half-rectified sine



$$x(t) = \begin{cases} E\sin\omega t & 0 < t < \pi/\omega \\ 0 & -\pi/\omega < t < 0 \end{cases}$$

$T = 2\pi/\omega, \quad \omega_1 = \omega$

Odd? No $\Rightarrow$ expect cosines

Even? No $\Rightarrow$ expect sines

DC component. Why does the integral stop at π/ω? $x = 0$ on $[\pi/\omega,\ 2\pi/\omega]$

$$A_0 = \frac{\omega}{\pi}\int_0^{\pi/\omega} E\sin\omega t\,dt = -\frac{E}{\pi}\big[\cos\pi - \cos 0\big] = \frac{2E}{\pi} \quad \Rightarrow \quad \text{DC} = \frac{A_0}{2} = \frac{E}{\pi} \approx 0.318E$$

Sine and cosine coefficients (only $0 < t < \pi/\omega$ contributes)

$$B_k = \frac{2}{T}\int_0^{\pi/\omega} E\sin(\omega t)\sin(k\omega t)\,dt = \frac{\omega E}{\pi}\int_0^{\pi/\omega} \sin(\omega t)\sin(k\omega t)\,dt$$

$k = 1$: $\int_0^{\pi/\omega}\sin^2(\omega t)\,dt = \frac{\pi}{2\omega} \Rightarrow B_1 = \frac{E}{2}$.

$k \neq 1$: $\sin\omega t\,\sin k\omega t = \frac{1}{2}[\cos(k-1)\omega t - \cos(k+1)\omega t]$, which integrates to terms in $\sin((k \mp 1)\pi) = 0$, so $B_k = 0$.

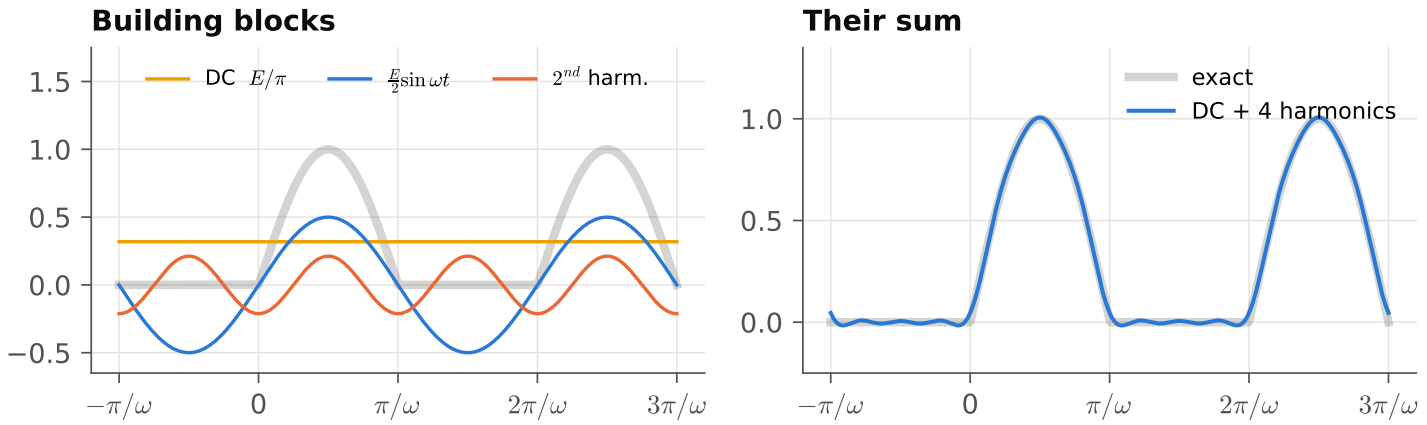
$$A_k = \frac{\omega E}{\pi}\int_0^{\pi/\omega} \sin(\omega t)\cos(k\omega t)\,dt, \text{ with } \sin\omega t\,\cos k\omega t = \tfrac{1}{2}[\sin(1+k)\omega t + \sin(1-k)\omega t]:$$

$A_1 = 0$, and for $k \neq 1$: $A_k = \dfrac{E\,(1 + (-1)^k)}{\pi(1 - k^2)}$

$\Rightarrow \ A_k = 0$ for odd k, $\qquad A_k = \dfrac{-2E}{\pi(k^2 - 1)}$ for even k

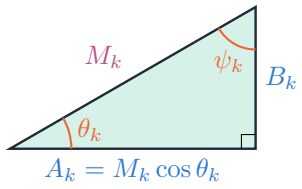
Series

$$x(t) = \frac{E}{\pi} + \frac{E}{2}\sin\omega t - \frac{2E}{\pi}\left[\frac{\cos 2\omega t}{3} + \frac{\cos 4\omega t}{15} + \frac{\cos 6\omega t}{35} + \cdots\right]$$



It is half of the original sine, plus a DC offset, plus even harmonics that flatten the negative half.

7 Magnitude–phase form (one-sided spectrum)



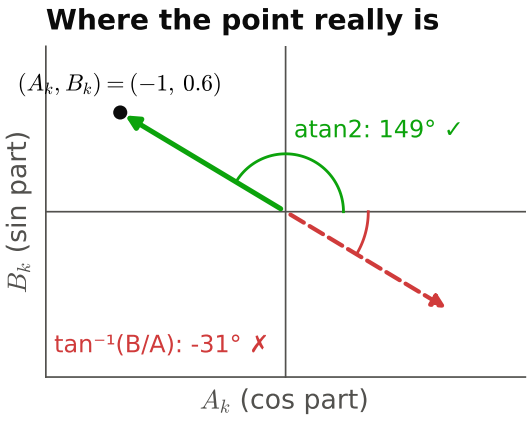
Expand $M_k\cos(k\omega_1 t - \theta_k)$ and match it with $A_k\cos + B_k\sin$:

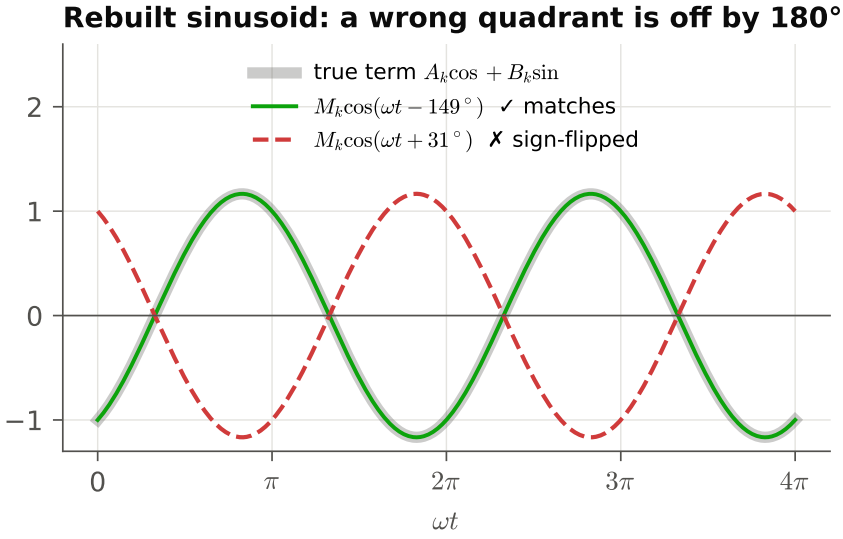
$$A_k = M_k\cos\theta_k, \quad B_k = M_k\sin\theta_k$$
$$M_k = \sqrt{A_k^2 + B_k^2}, \quad \theta_k = \tan^{-1}(B_k/A_k)$$

Sine form: $M_k\sin(k\omega_1 t + \psi_k)$, with $\psi_k = \tan^{-1}(A_k/B_k) = \frac{\pi}{2} - \theta_k$

$$x(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} M_k\cos(k\omega_1 t - \theta_k) = \frac{A_0}{2} + \sum_{k=1}^{\infty} M_k\sin(k\omega_1 t + \psi_k)$$

Watch out. $\tan^{-1}$ only returns angles in $(-\pi/2, \pi/2)$. Use the signs of A_k and B_k to pick the quadrant, i.e. $\theta_k = \text{atan2}(B_k, A_k)$. For example, $A_k < 0$ and $B_k = 0$ gives $\theta_k = \pi$, not 0.

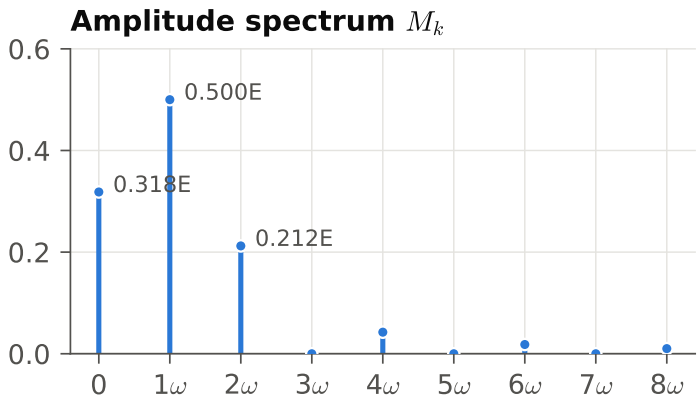


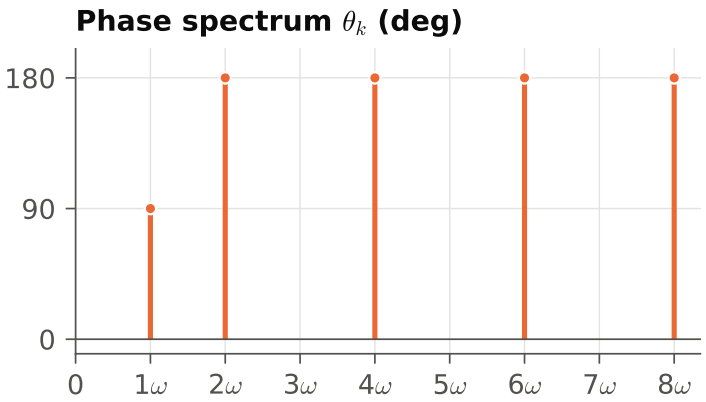


Example with $A_k = -1$, $B_k = 0.6$. A calculator's $\tan^{-1}(B_k/A_k)$ returns $-31°$, which points the opposite way (red). The rebuilt cosine is then the true term flipped upside down. $\text{atan2}(B_k, A_k) = 149°$ (green) reproduces the true term exactly.

7.1 Half-rectified sine in magnitude–phase form

k	0	1	2	3	4
A_k	$A_0/2 = E/\pi$	0	$-\frac{2E}{3\pi}$	0	$-\frac{2E}{15\pi}$
B_k	–	$E/2$	0	0	0
M_k	$0.318E$	$0.500E$	$0.212E$	0	$0.042E$
θ_k	–	$\pi/2$	π	–	π



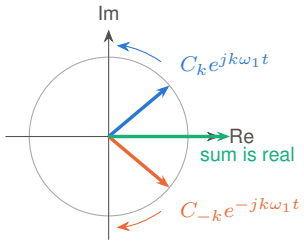


Amplitude falls like $1/k^2$. Most of the content is in the DC term, the fundamental and the 2nd harmonic. Phase is π for even harmonics because $A_k < 0$.

8 Complex exponential (Euler) form

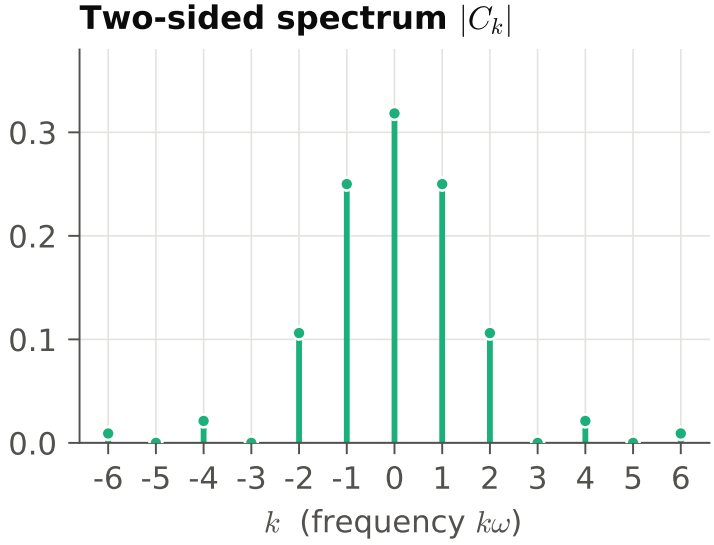
Use $\cos\phi = \frac{1}{2}(e^{j\phi} + e^{-j\phi})$ and $\sin\phi = \frac{1}{2j}(e^{j\phi} - e^{-j\phi})$:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k\, e^{jk\omega_1 t}, \qquad C_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jk\omega_1 t}\, dt$$



Each real sinusoid is a pair of counter-rotating phasors, each with half the amplitude. “Negative frequency” is bookkeeping.

$$C_0 = \tfrac{A_0}{2}$$
$$C_k = \tfrac{1}{2}(A_k - jB_k), \qquad C_{-k} = \tfrac{1}{2}(A_k + jB_k) = \overline{C_k}$$
$$|C_k| = \tfrac{1}{2}M_k, \qquad \angle C_k = \tan^{-1}\!\left(\tfrac{-B_k}{A_k}\right) = -\theta_k$$
$$A_k = 2\,\mathrm{Re}[C_k], \qquad B_k = -2\,\mathrm{Im}[C_k]$$



Half-rectified sine:

$C_0 = E/\pi$

$C_{\pm 1} = \mp jE/4$

$C_{\pm 2} = -E/(3\pi)$

The magnitude is symmetric in k and the phase is antisymmetric. The one-sided M_k is split in half between $\pm k$.

9 Summary: four equivalent forms

Form	Series	Coefficients
Sine–cosine	$\frac{A_0}{2} + \sum [A_k \cos k\omega_1 t + B_k \sin k\omega_1 t]$	compute by integration
Cosine series	$\frac{A_0}{2} + \sum M_k \cos(k\omega_1 t - \theta_k)$	$M_k = \sqrt{A_k^2 + B_k^2},\ \theta_k = \operatorname{atan2}(B_k, A_k)$
Sine series	$\frac{A_0}{2} + \sum M_k \sin(k\omega_1 t + \psi_k)$	$\psi_k = \operatorname{atan2}(A_k, B_k) = \frac{\pi}{2} - \theta_k$
Complex	$\sum_{-\infty}^{\infty} C_k e^{jk\omega_1 t}$	$C_k = \frac{1}{2}(A_k - jB_k),\ \lvert C_k\rvert = \frac{1}{2}M_k,\ \angle C_k = -\theta_k$

Back-conversion: $A_k = M_k \cos\theta_k = 2\,\mathrm{Re}[C_k], \quad B_k = M_k \sin\theta_k = -2\,\mathrm{Im}[C_k]$.

Exit question. $A_3 = -1$ and $B_3 = 1$. Find M_3, θ_3 and C_3.

$M_3 = \sqrt{2}, \quad \theta_3 = \operatorname{atan2}(1, -1) = 3\pi/4$ (*not* $-\pi/4$), $\quad C_3 = \frac{1}{2}(-1 - j)$, with $\lvert C_3\rvert = \frac{\sqrt{2}}{2}$ and $\angle C_3 = -3\pi/4$.

Teaching notes. Common mistakes: reporting A_0 instead of $A_0/2$ as the DC value; ignoring the quadrant in $\tan^{-1}$; integrating over less than a full period; mixing up rad/s and Hz. Slides 19–20 of the deck are hidden and not covered here. **Next lecture:** plot amplitude and phase spectra and combine them with $G(j\omega)$ to predict the response to periodic inputs (Section 1).